Exchange Rates

The sales prices for our ordinary shares traded on German stock exchanges are denominated in euro. Fluctuations in the exchange rate between the euro and the U.S. dollar affect the dollar equivalent of the euro price of the ordinary shares traded on the German stock exchanges and, as a result, may affect the price of the ADRs traded on the NYSE in the United States. See "*Item 9. The Offer and Listing*" for a description of the ADRs. In addition, SAP SE pays cash dividends, if any, in euro. As a result, any exchange rate fluctuations will also affect the dollar amounts received by the holders of ADRs on the conversion into dollars of cash dividends paid in euro on the ordinary shares represented by the ADRs. Deutsche Bank Trust Company Americas is the depositary (the Depositary) for SAP SE's ADR program. The deposit agreement with respect to the ADRs requires the Depositary to convert any dividend payments from euro into dollars as promptly as practicable upon receipt. For additional information on the Depositary and the fees associated with SAP's ADR program see "*Item 12. Description of Securities Other Than Equity Securities — American Depositary Shares*."

For details on the impact of exchange rate fluctuations see "*Item 5. Operating and Financial Review and Prospects — Foreign Currency Exchange Rate Exposure*".

Dividends

Dividend Distribution Policy

Dividends are jointly proposed by SAP SE's Supervisory Board (Aufsichtsrat) and Executive Board (Vorstand) based on SAP SE's year-end stand-alone statutory financial statements, subject to approval by the Annual General Meeting of Shareholders. Dividends are officially declared for the prior year at SAP SE's Annual General Meeting of Shareholders. SAP SE's Annual General Meeting of Shareholders usually convenes during the second quarter of each year. Beginning with the dividends payable for the 2017 fiscal year and in accordance with a change of the German Stock Corporation Act that aims to implement joint market standards in Europe for corporate actions processing, dividends are remitted to the custodian bank on behalf of the shareholders on the third business day following the Annual General Meeting of Shareholders. Record holders of the ADRs on the dividend record date will be entitled to receive payment of the dividend declared in respect of the year for which it is declared. Cash dividends payable to such holders will be paid to the Depositary in euro and, subject to certain exceptions, will be converted by the Depositary into U.S. dollars.

Dividends paid to holders of the ADRs may be subject to German withholding tax. See "*Item 8. Financial Information — Other Financial Information — Dividend Policy*" and "*Item 10. Additional Information — Taxation*," for further information.

Annual Dividends Paid and Proposed

The following table sets forth in euro the annual dividends paid or proposed to be paid per ordinary share in respect of each of the years indicated. One SAP ADR currently represents one SAP SE ordinary share. Accordingly, the final dividend per ADR is equal to the dividend for one SAP SE ordinary share and is dependent on the euro/U.S. dollar exchange rate. The table does not reflect tax credits that may be available to German taxpayers who receive dividend payments. If you own our ordinary shares or ADRs and if you are a U.S. resident, refer to "*Item 10. Additional Information — Taxation,*" for further information.

	Dividend Paid per Ordinary Share	
Year Ended December 31,	**€**	**US$**
2015	1.15	1.30 [1]
2016	1.25	1.37 [1]
2017	1.40	1.65 [1]
2018	1.50	1.68 [1]
2019 (proposed)	1.58 [2]	1.73 [2], [3]

[1] Translated for the convenience of the reader from euro into U.S. dollars at the Noon Buying Rate for converting euro into U.S. dollars on the dividend payment date. The Depositary is required to convert any dividend payments received from SAP as promptly as practicable upon receipt.

[2] Subject to approval at the Annual General Meeting of Shareholders of SAP SE currently scheduled to be held on May 15, 2020.

[3] Translated for the convenience of the reader from euro into U.S. dollars at the Noon Buying Rate for converting euro into U.S. dollars on February 7, 2020 of US$1.0950 per €1.00. The dividend paid may differ due to changes in the exchange rate.

The amount of dividends paid on the ordinary shares depends on the amount of profits to be distributed by SAP SE, which depends in part upon our financial performance. In addition, the amount of dividends received by holders of ADRs may be affected by fluctuations in exchange rates (see "*Item 3. Key Information — Exchange Rates*"). The timing, declaration, amount and payment of any future dividend will depend upon our future earnings, capital needs and other relevant factors, in each case as proposed by the Executive Board and the Supervisory Board of SAP SE and approved by the Annual General Meeting of Shareholders.

In light of SAP's strong financial performance and healthy balance sheet, the Supervisory Board of SAP SE approved, on November 4, 2019, the Executive Board's plan for enhanced capital return in 2020. Under this new program, the Company intends to repurchase shares and/or issue a special dividend with a combined volume of €1.5 billion by December 31, 2020.

$0.30 in 2019 as a result of the adoption of the new revenue recognition standard. Adjusted EPS is a non-GAAP financial measure.

SEGMENT RESULTS

Cubic Transportation Systems

	Fiscal 2019	Fiscal 2018	% Change
	(in millions)		
Sales	$ 849.8	$ 670.7	27 %
Operating income	77.2	60.4	28
Adjusted EBITDA	110.5	73.3	51

Sales: CTS sales increased 27% to $849.8 million in 2019 compared to $670.7 million in 2018, including the impact of the adoption of ASC 606. The increase in sales was primarily driven by growth in both organic and inorganic business in North America. Sales in 2019 were higher in the U.S. primarily due to system development on contracts in New York, Boston, and the San Francisco Bay Area. Businesses acquired by CTS during fiscal year 2019, whose operations are all located in the U.S., had sales of $74.4 million in fiscal year 2019. Sales increased slightly in Australia between fiscal years 2018 and 2019 as increased system development work on a contract in Brisbane was partially offset by the negative impact of foreign currency exchange rates as well as a decrease in service sales. Sales were lower in the UK primarily due to a decrease in system development work in London and the negative impact of currency exchange rates. The average exchange rates between the prevailing currencies in our foreign operations and the U.S. dollar resulted in a decrease in CTS sales of $22.2 million for 2019 compared to 2018, primarily due to the strengthening of the U.S. dollar against the British pound and Australian dollar.

Amortization of Purchased Intangibles: Amortization of purchased intangibles included in the CTS operating results totaled $22.0 million in 2019 and $5.2 million in 2018. The increase is due to the amortization of purchased intangibles for companies acquired by CTS in fiscal year 2019.

Operating Income: CTS operating income increased 28% in 2019 to $77.2 million compared to $60.4 million in 2018. The increase in operating income was primarily caused by higher margins on increased work on development projects in New York, Boston, the San Francisco Bay Area and Brisbane, as well as the impact of the adoption of ASC 606. These increases in operating income were partially offset by operating losses incurred by businesses acquired by CTS in fiscal 2019 as well as the negative impact of changes in foreign currency exchange rates. Businesses acquired by CTS in fiscal years 2019 incurred operating losses of $10.1 million in fiscal 2019, which included acquisition transaction costs of $8.1 million and amortization of intangible assets totaling $19.3 million. The average exchange rates between the prevailing currency in our foreign operations and the U.S. dollar resulted in a decrease in CTS operating income of $3.6 million for 2019 compared to 2018.

Adjusted EBITDA: CTS Adjusted EBITDA increased 51% to $110.5 million in 2019 compared to $73.3 million in 2018. The increase in Adjusted EBITDA was primarily driven by the same factors that drove the increase in operating income described above excluding the increases in amortization of purchased intangibles and acquisition transaction costs which are excluded from Adjusted EBITDA. Adjusted EBITDA for CTS increased by $2.3 million in 2019 as a result of the adoption of the new revenue recognition standard.

Net Sales

Solid Capacitors net sales of $771.2 million in fiscal year 2018 increased $196.1 million or 34.1% from $575.1 million in fiscal year 2017. Tantalum product line net sales of $495.1 million in fiscal year 2018 increased $152.9 million or 44.7% from $342.2 million in fiscal year 2017. Ceramic product line net sales of $276.1 million in fiscal year 2018 increased $43.2 million or 18.5% from $232.9 million in fiscal year 2017.

The increase in Solid Capacitors net sales was primarily driven by the addition of net sales of $133.8 million resulting from the TOKIN acquisition and an increase in net sales to the legacy products distributor channel of $81.7 million. To a lesser degree, an increase in legacy Ceramic products' net sales of $6.0 million in the EMS channel across all regions and $10.2 million in the OEM channel in the EMEA and APAC regions also contributed to the increase in Solid Capacitors net sales. These increases were partially offset by a $28.0 million decrease in net sales in the OEM channel for legacy Tantalum products across all regions. In addition, Solid Capacitors net sales was favorably impacted by $6.1 million from foreign currency exchange due to the change in the value of the Euro compared to the U.S. dollar.

Reportable Segment Operating Income

Segment operating income of $234.5 million for fiscal year 2018 increased $86.8 million or 58.8% from $147.7 million for fiscal year 2017. The increase in operating income was primarily attributable to an increase in gross margin of $100.4 million. TOKIN contributed $47.0 million in additional gross margin in fiscal year 2018. Legacy KEMET gross margin increased $53.3 million, or 30.6%, primarily driven by an increase in net sales, cost improvements in vertical integration, favorable foreign currency impact to manufacturing costs, and manufacturing process improvements resulting from our cost reduction activities. In addition, there was a $1.6 million improvement in (gain) loss on write down and disposal of long-lived assets. Partially offsetting these improvements were a $10.5 million increase in SG&A expenses and a $5.0 million increase in R&D expenses. TOKIN accounted for $10.2 million of the increase in SG&A expenses and $4.1 million of the increase in R&D expenses.

Film and Electrolytic

The table below sets forth net sales, operating income (loss) and operating income (loss) as a percentage of net sales for our Film and Electrolytic reportable segment for the fiscal years 2018 and 2017 (amounts in thousands, except percentages):

	For the Fiscal Years Ended			
	March 31, 2018		March 31, 2017	
	Amount	% to Net Sales	Amount	% to Net Sales
Net sales [(1)]	$ 201,977		$ 182,228	
Segment operating income (loss) [(1)]	3,622	1.8%	(9,028)	(5.0)%

[(1)] *Fiscal years ending March 31, 2018 and 2017 adjusted due to the adoption of ASC 606*

Net Sales

Film and Electrolytic net sales of $202.0 million in fiscal year 2018 increased $19.7 million or 10.8% from $182.2 million in fiscal year 2017. The increase in net sales was primarily driven by an increase in net sales in the distributor channel across all the APAC and EMEA regions of $13.7 million, a $3.3 million increase in net sales in the OEM channel of the EMEA region and a $4.2 million increase in the EMS channel across the Americas, EMEA, and APAC regions. These increases were partially offset by a decrease in net sales of $1.2 million in the OEM channel across the Americas, APAC, and JPKO regions. In addition, there was a favorable impact of $7.6 million from foreign currency exchange primarily due to the change in the value of the Euro compared to the U.S. dollar.

Reportable Segment Operating Income (Loss)

Segment operating income of $3.6 million in fiscal year 2018 improved $12.7 million from $9.0 million of operating loss in fiscal year 2017. The improvement was primarily attributable to a $4.3 million increase in gross margin driven by higher net sales, as well as the benefit of completed restructuring activities. The increase was also attributed to an $11.7 million improvement in (gain) loss on the write down and disposal of long-lived assets. These improvements were partially offset by a $2.1 million increase in restructuring charges, a $0.7 million increase in SG&A expenses, and a $0.6 million increase in R&D expenses.

NAVIOS MARITIME HOLDINGS INC.
NOTES TO THE CONSOLIDATED FINANCIAL STATEMENTS
(Expressed in thousands of U.S. dollars — except share data)

NOTE 5: ACCOUNTS RECEIVABLE, NET

Accounts receivable consisted of the following:

	December 31, 2019	December 31, 2018
Accounts receivable	$ 68,309	$ 76,376
Less: provision for doubtful receivables	(16,377)	(16,086)
Accounts receivable, net	**$ 51,932**	**$ 60,290**

Changes to the provisions for doubtful accounts are summarized as follows:

Allowance for doubtful receivables	Balance at Beginning of Period	Charges to Costs and Expenses	Amount Utilized	Balance at End of Period
Year ended December 31, 2017	$ (19,437)	$ (269)	$ —	$(19,706)
Year ended December 31, 2018	$ (19,706)	$ (575)	$4,195	$(16,086)
Year ended December 31, 2019	$ (16,086)	$ (999)	$ 708	$(16,377)

Concentration of credit risk with respect to accounts receivable is limited due to the Company's large number of customers, who are internationally dispersed and have a variety of end markets in which they sell. Due to these factors, management believes that no additional credit risk beyond amounts provided for collection losses is inherent in the Company's trade receivables. For the year ended December 31, 2019, two customers accounted for 17.1% and 14.4%, respectively, of the Company's revenue from continuing operations and are the same customers who accounted for 13.1% and 11.6%, respectively, of the Company's revenue from continuing operations in the year ended December 31, 2018. For the year ended December 31, 2017, no customers accounted for more than 10% of the Company's revenue.

NOTE 6: PREPAID EXPENSES AND OTHER CURRENT ASSETS

Prepaid expenses and other current assets consisted of the following:

	December 31, 2019	December 31, 2018
Prepaid voyage and operating costs	$ 5,726	$ 9,261
Claims receivable	3,826	22,224
Prepaid other taxes	1,012	2,682
Advances for working capital purposes	—	18
Other	1,675	6,005
Total prepaid expenses and other current assets	**$ 12,239**	**$ 40,190**

Claims receivable mainly represents claims against vessels' insurance underwriters in respect of damages arising from accidents or other insured risks, as well as claims under charter contracts including off-hires. While it is anticipated that claims receivable will be recovered within one year, such claims may not all be recovered within one year due to the attendant process of settlement. Nonetheless, amounts are classified as current as they represent amounts currently due to the Company. All amounts are shown net of applicable deductibles.

As of December 31, 2018, claims receivable include $11,571 related to insurance claim at the iron ore port terminal in Nueva Palmira, Uruguay.

NOTE 7: VESSELS, PORT TERMINALS AND OTHER FIXED ASSETS, NET

Vessels	Cost	Accumulated Depreciation	Net Book Value
Balance December 31, 2016	**$1,930,950**	**$ (521,535)**	**$1,409,415**
Additions	—	(73,017)	(73,017)
Impairment losses	(104,157)	58,034	(46,123)
Disposals	(11,828)	—	(11,828)
Balance December 31, 2017	**1,814,965**	**(536,518)**	**1,278,447**

The Company's operations in the Netherlands, Belgium and the United States receive research and development grants and credits from various sources.

Personnel expenses for employees were as follows:

	December 31,	
	2018	2019
Wages and salaries	158,371	**191,459**
Social security	14,802	**17,214**
Pension expenses	6,937	**8,408**
Share-based payment expenses	8,215	**10,538**
Restructuring expenses	178	**108**
Total	**188,503**	**227,727**

Personnel expenses are included in cost of sales and in operating expenses in the consolidated statement of profit or loss.

The number of employees, exclusive of temporary workers, by geographical area at year-end was as follows:

	December 31,	
Geographical location	2018	2019
Europe:		
- the Netherlands	151	**145**
- EMEA	189	**203**
United States	576	**639**
Japan	248	**271**
South Korea	273	**280**
Singapore	463	**474**
Asia, other	281	**325**
Total	**2,181**	**2,337**

The number of employees, exclusive of temporary workers, by function at year-end was as follows:

	December 31,	
Per function	2018	2019
Research and development	544	**612**
Manufacturing	456	**484**
Marketing and sales	277	**275**
Customer service	716	**779**
Finance and administration	188	**187**
Total	**2,181**	**2,337**

NOTE 24. EARNINGS PER SHARE

Basic net earnings per common share is calculated by dividing net income attributable to common shareholders by the weighted average number of common shares outstanding for that period. The dilutive effect is calculated using the treasury stock method. The calculation of diluted net income per share assumes the exercise of options issued under our stock option plans (and the issuance of shares under our share plans) for periods in which exercises (or issuances) would have a dilutive effect.

The calculation of basic and diluted net income per share attributable to common shareholders is based on the following data:

	December 31,	
	2018	2019
Net earnings used for purposes of calculating net income per common share		
Net earnings from operations	**157,133**	**329,013**
Basic weighted average number of shares outstanding during the year (thousands)	52,432	**49,418**
Effect of dilutive potential common shares from stock options and restricted shares	678	**580**
Dilutive weighted average number of shares outstanding	**53,110**	**49,999**
Basic net earnings per share:		
from operations	3.00	**6.66**
Diluted net earnings per share:		
from operations	2.96	**6.58**

€254million – from the reported balance sheet profit of €266 million and to carry forward the remaining amount to the new account.

32. Provisions for post-employment benefits plans and similar obligations

€ million	30/9/2018	**30/9/2019**
Provisions for post-employment benefits plans (employer's commitments)	344	**414**
Provisions for indirect commitments	12	**17**
Provisions for voluntary pension benefits	0	**0**
Provisions for post-employment benefit plans	71	**78**
	427	**509**
Provisions for obligations similar to pensions	41	**34**
	468	**543**

Provisions for post-employment benefits plans are recognised in accordance with IAS 19 (Employee Benefits).

Provisions for post-employment benefits plans consist of commitments primarily related to benefits defined by the provisions of company pension plans. These take the form of defined benefit plans directly from the employer (employer's commitments) and defined benefit plans from external pension providers (benevolent funds in Germany and international pension funds). The external providers' assets serve exclusively to finance the pension entitlements and qualify as plan assets. The benefits under the different plans are based on performance and length of service.

The most important performance-based pension plans are described in the following.

Germany

METRO grants many employees in Germany retirement, disability and surviving dependant's benefits. New commitments are granted in the form of 'defined benefit' commitments in the meaning of IAS 19 (contribution-oriented commitments pursuant to German company pension law), which comprise a payment contribution component and an employer-matching component. Contributions are paid to a pension reinsurance from which benefits are paid out when the insured event occurs. A provision is recognised for entitlements not covered by pension reinsurance.

In addition, various pension funds exist that are closed for new contributions. In general, these provide for lifelong pensions starting with the start of retirement or recognised invalidity. Benefits are largely defined as fixed payments or on the basis of set annual increases. In special cases, benefits are calculated in consideration of accrued statutory pension entitlements. The commitments provide for a widow's or widower's pension of varying size, depending on the benefits the former employee received or would have received in case of invalidity. Legacy commitments are partially covered by assets held in benevolent funds. Provisions are recognised for those commitments not covered. The benevolent funds' decision-making bodies (management board and general assembly of members) comprise both employer and employee representatives. The respective members of the Management Board decide on the deployment of funds and financial investments. It may commission third parties to manage fund assets. No statutory minimum endowment

During the year ended 30 June 2019, the Group contributed a further $1.3m into the Bundall Storage Trust and $2.2m into the Bundall Commercial Trust as part of a capital raise. There was no change in the total share of the Group's interest in either Trust following this investment.

On 21 June 2019, the Group purchased two storage centre investment property assets from the Bundall Storage Trust for $43.7m.

On 21 June 2019, the Bundall Storage Trust purchased a site for a proposed storage centre development from the Group for $8.2m and the Bundall Commercial Trust purchased a commercial property from the Group for $17.8m.

As at 30 June 2019 the Bundall Storage Trust has one storage centre investment property asset under construction. The Bundall Commercial Trust derives rental property income from the leasing of commercial units.

During the year ended 30 June 2018, the Group subscribed to 83.3% of the units in FKS Investments No.2 Unit Trust ("**FKS**"). FKS subsequently purchased a storage centre investment property asset in Queensland, Australia. On 28 June 2019, the Group sold its units in FKS for $3m.

These investments are classified as joint ventures as all parties are subject to a Securityholders Agreement that has been contractually structured such that the parties to the agreement have equal representation on the advisory board responsible for the overall direction and supervision of each trust.

None of the Group's joint ventures are listed on any public exchange.

None of the Group's other joint ventures have any capital commitments at 30 June 2019. None of the Group's joint ventures had any contingent liabilities at 30 June 2019.

Interest in associates

	2019 $'000	2018 $'000
Opening balance at 1 July	10,693	8,611
Capital contribution / acquisition of shareholding in associates	-	2,048
Share of profit from associates*	1,695	1,282
Distributions from associate	-	(1,248)
Closing balance at 30 June	12,388	10,693

**Included within share of profit from associates is $1,917,000 representing NSR's share of fair value gains related to investment properties held by joint ventures and associates (30 June 2018: $1,383,000).*

The Group owns 24.9% (2018: 24.9%) of the Australia Prime Storage Fund ("**APSF**"). APSF is a partnership with Universal Self Storage to facilitate the development and ownership of multiple premium grade self-storage centres in select cities around Australia.

During the year ended 30 June 2019, National Storage (Operations) Pty Ltd earned fees of $0.8m from APSF associated with the design, development, financing of the construction process, and ongoing management of centres (see note 17) (30 June 2018: $0.7m).

As at 30 June 2019, APSF had two operating centres in Queensland, Australia, with a third asset under construction in Victoria, Australia.

Following the financial year end, on 26 July 2019, the Group purchased two storage centre investment properties from APSF for $42.6m, and reached an agreement to purchase a third asset for $21.35m on completion of construction (see note 23). During the year ended 30 June 2018, the Group purchased a storage centre investment property asset in Queensland, Australia from APSF for $14m.

As at 30 June 2019, APSF had contractual commitments of $2.8m in place for the construction of one storage centre in Victoria, Australia. Neither associate had any contingent liabilities or any other capital commitments at 30 June 2019 or 30 June 2018.

The Group holds a 24% (30 June 2018: 24.8%) holding in Spacer Marketplaces Pty Ltd ("**Spacer**"). Spacer operate online peer-to-peer marketplaces for self-storage and parking.

13. CONTRIBUTED EQUITY

	2019 $'000	2018 $'000
Issued and paid up capital	83,692	66,128
Contract for future issue of equity	16,451	-
Total contributed equity	100,143	66,128

Number of stapled securities on Issue	2019	2018
Opening balance at 1 July	559,107,042	512,913,914
Institutional and retail placement	105,677,937	39,712,882
Distribution reinvestment plan	9,143,772	6,480,246
Closing balance at 30 June	673,928,751	559,107,042

Capital raise
On 4 September 2018, the Group undertook a fully underwritten $175.4m equity raising. This resulted in the issue of 105,677,937 new stapled securities (2018: $59.5m equity raising resulting in the issue of 39,712,882 stapled securities).

On 25 June 2019, the Group announced a fully underwritten $170m equity raising. On 28 June 2019, the Group received proceeds for this raising. This has been recognised as a contract for future issue of equity under AASB 132 and has been recognised as contributed equity within the statement of financial position. This resulted in the issue of 99,415,205 new stapled securities on 1 July 2019. These securities are not reflected in the securities on issue above as they were issued subsequent to the year end.

On 25 June 2019, the Group also announced a non-underwritten security purchase plan. This completed on 30 July 2019, raising $13.5m and resulted in the issue of 7,917,735 new stapled securities.

Distribution reinvestment plan
During the year, 9,143,772 (2018: 6,480,246) stapled securities were issued to securityholders participating in the Group's Distribution Reinvestment Plan for consideration of $16.2m (2018: $9.6m). The stapled securities were issued at the volume weighted average market price of the Group's stapled securities over a period of ten trading days, less a 2% discount.

Terms and conditions of contributed equity

Stapled securities
A stapled security represents one share in NSH and one unit in NSPT. Stapled securityholders have the right to receive declared dividends from NSH and distributions from NSPT and are entitled to one vote per stapled security at securityholders' meetings. Holders of stapled securities can vote their shares and units in accordance with the Corporations Act 2001, either in person or by proxy, at a meeting of either NSH or NSPT. The stapled securities have no par value.

In the event of the winding up of NSH and NSPT, stapled securityholders have the right to participate in the proceeds from the sale of all surplus assets in proportion to the number of and amounts paid up on stapled securities held. Ordinary stapled securityholders rank after all creditors in repayment of capital.

There is no current on or off market security buy-back.

Other gains, net. We recorded net other gains totalling RMB19,689 million for the year ended 31 December 2019, which primarily comprised of non-IFRS adjustment items such as fair value gains arising from increased valuations for certain investee companies in verticals such as FinTech services, social media and education, as well as net deemed disposal gains arising from the capital activities of certain investee companies in verticals including transportation services and online games.

Selling and marketing expenses. Selling and marketing expenses decreased by 12% to RMB21,396 million for the year ended 31 December 2019 on a year-on-year basis. The decrease was mainly due to the reduction of advertising and promotion expenses as a result of improved operational efficiencies. As a percentage of revenues, selling and marketing expenses decreased to 6% for the year ended 31 December 2019 from 8% for the year ended 31 December 2018.

General and administrative expenses. General and administrative expenses increased by 29% to RMB53,446 million for the year ended 31 December 2019 on a year-on-year basis. The increase was primarily driven by greater R&D expenses and staff costs. As a percentage of revenues, general and administrative expenses increased to 14% for the year ended 31 December 2019 from 13% for the year ended 31 December 2018.

Finance costs, net. Net finance costs increased by 63% to RMB7,613 million for the year ended 31 December 2019 on a year-on-year basis. The increase primarily reflected greater interest expenses resulting from higher amounts of indebtedness.

Share of (loss)/profit of associates and joint ventures. We recorded share of losses of associates and joint ventures of RMB1,681 million for the year ended 31 December 2019, compared to share of profit of RMB1,487 million for the year ended 31 December 2018. The change was mainly due to non-cash charges booked by certain associates.

Income tax expense. Income tax expense decreased by 7% to RMB13,512 million for the year ended 31 December 2019 on a year-on-year basis. The decrease mainly reflected the entitlements of preferential tax treatments and benefits.

Profit attributable to equity holders of the Company. Profit attributable to equity holders of the Company increased by 19% to RMB93,310 million for the year ended 31 December 2019 on a year-on-year basis. Non-IFRS profit attributable to equity holders of the Company increased by 22% to RMB94,351 million for the year ended 31 December 2019.

The deferred tax assets and liabilities result from differences in the timing of the recognition of certain income and expense items for tax and financial accounting purposes. The sources of these differences at each balance sheet date are as follows (in thousands):

	December 31,	
	2019	2018
Deferred income tax assets:		
Net operating loss carryforwards	$ 23,030	$ 25,745
Tax credits	52,902	43,838
Compensation	18,791	15,934
Deferred revenue	25,599	27,587
Research and development expense deferral	—	12,631
Other	4,065	5,393
Gross deferred income tax assets	124,387	131,128
Less: valuation allowance	(7,653)	(20,415)
Net deferred income tax assets	$ 116,734	$ 110,713
Deferred income tax liabilities:		
Depreciation and amortization	$ (52,978)	$ (60,872)
Deferred revenue	(44,198)	(54,508)
Total deferred income tax liabilities	(97,176)	(115,380)
Net deferred income taxes	$ 19,558	$ (4,667)
Deferred income taxes / liabilities included in the balance sheet are:		
Deferred income tax asset – noncurrent	$ 51,611	$ 27,048
Deferred income tax liability – noncurrent	(32,053)	(31,715)
Net deferred income taxes	$ 19,558	$ (4,667)

In assessing the realizability of deferred tax assets, the Company considers whether it is more likely than not that some portion or all of the deferred tax assets will not be realized. The ultimate realization of deferred tax assets is dependent upon the generation of future taxable income during the periods in which those temporary differences become deductible. The Company considers projected future taxable income, carryback opportunities, and tax planning strategies in making this assessment. Based upon the level of historical taxable income and projections for future taxable income over the periods which the deferred tax assets are deductible, the Company believes it is more likely than not that it will realize the benefits of these deductible differences, net of the valuation allowances recorded. During the year ended December 31, 2019, the Company decreased its valuation allowance by $12.8 million which relates to a reduction in the valuation allowance on U.S. foreign tax credits offset by an increase in valuation allowance on foreign net operating losses.

At December 31, 2019, the Company had domestic federal tax net operating losses ("NOLs") of $65.9 million, which will begin to expire in 2020. The Company had deferred tax assets equal to $1.4 million related to domestic state tax NOLs which will begin to expire in 2020. The Company does not have any valuation allowance against the federal tax NOLs but has provided a $1.2 million valuation allowance against the deferred tax asset associated with the state NOLs. The Company had foreign tax NOLs of $30.4 million, of which $28.1 million may be utilized over an indefinite life, with the remainder expiring over the next 17 years. The Company has provided a $0.7 million valuation allowance against the deferred tax asset associated with the foreign NOLs.

The Company had U.S. foreign tax credit carryforwards at December 31, 2019, of $40.7 million, for which an $1.2 million valuation allowance has been provided. The U.S. foreign tax credits will begin to expire in 2022. The Company had foreign tax credit carryforwards in other foreign jurisdictions at December 31, 2019, of $1.9 million, of which $1.3 million may be utilized over an indefinite life, with the remainder expiring over the next seven years. The Company has provided a $1.2 million valuation allowance against the tax benefit associated with these foreign credits. The Company also has domestic federal and state general business tax credit carryforwards at December 31, 2019, of $15.7 million and $0.8 million, respectively, which will begin to expire in 2020 and 2022, respectively.

The unrecognized tax benefit at December 31, 2019 and 2018, was $29.0 million and $28.4 million, respectively, of which $22.4 million and $22.6 million, respectively, are included in other noncurrent liabilities in the consolidated balance sheets. Of the total

LIQUIDITY AND CAPITAL RESOURCES

This data should be read in conjunction with our Consolidated Statements of Cash Flows.

(in millions)	As of November 29, 2019	As of November 30, 2018
Cash and cash equivalents	$ 2,650.2	$ 1,642.8
Short-term investments	$ 1,526.8	$ 1,586.2
Working capital	$ (1,696.0)	$ 555.9
Stockholders' equity	$ 10,530.2	$ 9,362.1

Working Capital

Working capital as of November 29, 2019 and November 30, 2018 was $1.70 billion of a deficit and $555.9 million of a surplus, respectively. The decrease was primarily due to the reclassification of $3.15 billion total carrying value of our $2.25 billion term loan due April 30, 2020 ("Term Loan") and $900 million 4.75% senior notes due February 1, 2020 ("2020 Notes") to current liabilities. We intend to refinance our Term Loan and 2020 Notes on or before the due dates.

A summary of our cash flows is as follows:

(in millions)	2019	2018	2017
Net cash provided by operating activities	$ 4,421.8	$ 4,029.3	$ 2,912.9
Net cash used for investing activities	(455.6)	(4,685.3)	(442.9)
Net cash used for financing activities	(2,946.1)	(5.6)	(1,183.7)
Effect of foreign currency exchange rates on cash and cash equivalents	(12.7)	(1.7)	8.5
Net increase (decrease) in cash and cash equivalents	$ 1,007.4	$ (663.3)	$ 1,294.8

Our primary source of cash is receipts from revenue and, to a lesser extent, proceeds from participation in the employee stock purchase plan. The primary uses of cash are our stock repurchase program as described below, payroll-related expenses, general operating expenses including marketing, travel and office rent, and cost of revenue. Other uses of cash include business acquisitions, purchases of property and equipment and payments for taxes related to net share settlement of equity awards.

Cash Flows from Operating Activities

For fiscal 2019, net cash provided by operating activities of $4.42 billion was primarily comprised of net income adjusted for the net effect of non-cash items. The primary working capital sources of cash were net income coupled with an increase in deferred revenue, which was offset in large part by cash outflows due to an increase in prepaid expenses and other assets. The increase in deferred revenue was primarily driven by increases related to Digital Media offerings with cloud-enabled services and Digital Experience hosted services. The primary working capital use of cash was due to increases in prepaid expenses with certain vendors, sales commissions paid and capitalized, advanced payments related to income taxes and increase in long-term contract assets.

Cash Flows from Investing Activities

For fiscal 2019, net cash used for investing activities of $455.6 million was primarily due to purchases of property and equipment and our acquisition of the remaining equity interest in Allegorithmic. These cash outflows were offset primarily by proceeds from sales and maturities of short-term investments, net of purchases. *See Note 3 of our Notes to Consolidated Financial Statements for more detailed information regarding our acquisitions.*

Cash Flows from Financing Activities

For fiscal 2019, net cash used for financing activities was $2.95 billion primarily due to payments for our treasury stock repurchases and taxes related to net share settlement of equity awards, which were offset by proceeds from re-issuance of treasury stock for our employee stock purchase plan. *See the section titled "Stock Repurchase Program" discussed below.*

We expect to continue our investing activities, including short-term and long-term investments, facilities expansion and purchases of computer systems for research and development, sales and marketing, product support and administrative staff. Furthermore, cash reserves may be used to repurchase stock under our stock repurchase program and to strategically acquire companies, products or technologies that are complementary to our business.

On a constant currency basis, acquisition related and other expenses decreased in fiscal 2019 compared to fiscal 2018 primarily due to certain favorable business combination related adjustments that were recorded in fiscal 201 9 .

Restructuring Expenses: Restructuring expenses resulted from the execution of management approved restructuring plans that were generally developed to improve our cost structure and/or operations, often in conjunction with our acquisition integration strategies. Restructuring expenses consist of employee severance costs and may also include charges for duplicate facilities and other contract termination costs to improve our cost structure prospectively. For additional information regarding our restructuring plans, see Note 8 of Notes to Consolidated Financial Statements included elsewhere in this Annual Report.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Restructuring expenses	$ 443	-25%	-22%	$ 588

Restructuring expenses in fiscal 2019 primarily related to our 2019 Restructuring Plan. Restructuring expenses in fiscal 2018 primarily related to our 2017 Restructuring Plan, which is substantially complete. Our management approved, committed to and initiated these plans in order to restructure and further improve efficiencies in our operations. In the fourth quarter of fiscal 2019, our management supplemented the 2019 Restructuring Plan to reflect additional actions that we expect to take. The total estimated restructuring costs associated with the 2019 Restructuring Plan are up to $584 million, of which approximately $108 million remained as of May 31, 2019, and will be recorded to the restructuring expense line item within our consolidated statements of operations as the costs are incurred through an expected end date during fiscal 2020. Our estimated costs are subject to change in future periods. We may incur additional restructuring expenses in future periods due to the initiation of new restructuring plans or from changes in estimated costs associated with existing restructuring plans .

Interest Expense:

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Interest expense	$ 2,082	3%	3%	$ 2,025

Interest expense increased in fiscal 2019 compared to fiscal 2018 primarily due to higher average borrowings resulting from our issuance of $10.0 billion of senior notes in November 2017, which was partially offset by a reduction in interest expense resulting primarily from the maturities and repayments of $2.0 billion of senior notes during fiscal 2019 and $6.0 billion of senior notes during fiscal 2018.

Non-Operating Income, net: Non-operating income, net consists primarily of interest income, net foreign currency exchange losses, the noncontrolling interests in the net profits of our majority-owned subsidiaries (primarily Oracle Financial Services Software Limited and Oracle Corporation Japan) and net other income, including net recognized gains and losses related to all of our investments, net unrealized gains and losses related to the small portion of our investment portfolio related to our deferred compensation plan, net unrealized gains and losses related to certain equity securities and non-service net periodic pension income (losses).

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Interest income	$ 1,092	-9%	-9%	$ 1,203
Foreign currency losses, net	(111)	50%	62%	(74)
Noncontrolling interests in income	(152)	12%	12%	(135)
Other income, net	(14)	-107%	-42%	191
Total non-operating income, net	$ 815	-31%	-31%	$ 1,185

Net Periodic Benefit Cost

The following table provides information about the net periodic benefit cost for the plans for fiscal years 2019, 2018 and 2017 (in thousands):

	Pension		
	2019	**2018**	**2017**
Service cost	$ 1,437	$ 1,063	$ 1,068
Interest cost	3,715	3,807	2,942
Expected long-term return on plan assets	(5,291)	(5,954)	(4,206)
Recognized actuarial loss	741	1,127	1,929
Amortization of prior service credit	(44)	(88)	(138)
Net settlement loss	634	116	1,472
Net periodic benefit cost	$ 1,192	$ 71	$ 3,067

On September 1, 2018, the Company adopted a new accounting standard, which changes the presentation of net periodic benefit cost in the Consolidated Statements of Operation. The Company adopted the standard on a retrospective basis which results in reclassifications for the service cost component of net periodic benefit cost from selling, general and administrative expense to cost of revenue and for the other components from selling, general and administrative expense to other expense. Prior periods have not been reclassified due to immateriality.

Assumptions

Weighted-average actuarial assumptions used to determine net periodic benefit cost and projected benefit obligation for the plans for the fiscal years 2019, 2018 and 2017 were as follows:

	Pension		
	2019	**2018**	**2017**
Net periodic benefit cost:			
Expected long-term return on plan assets[1]	3.6%	3.8%	3.3%
Rate of compensation increase	4.4%	3.3%	2.7%
Discount rate	2.2%	2.1%	1.9%
Projected benefit obligation:			
Expected long-term return on plan assets	2.0%	3.6%	4.0%
Rate of compensation increase	4.3%	4.4%	4.4%
Discount rate[2]	1.7%	2.2%	2.3%

(1) The expected return on plan assets assumption used in calculating net periodic benefit cost is based on historical return experience and estimates of future long-term performance with consideration to the expected investment mix of the plan.

(2) The discount rate is used to state expected cash flows relating to future benefits at a present value on the measurement date. This rate represents the market rate for high-quality fixed income investments whose timing would match the cash outflow of retirement benefits. Other assumptions include demographic factors such as retirement, mortality and turnover.

Plan Assets

The Company has adopted an investment policy for a majority of plan assets, which was set by plan trustees who have the responsibility for making investment decisions related to the plan assets. The plan trustees oversee the investment allocation, including selecting professional investment managers and setting strategic targets. The investment objectives for the assets are (1) to acquire suitable assets that hold the appropriate liquidity in order to

The following table details the Company's sales by operating segment for fiscal years ended September 30, 2019 and 2018. The Company's sales by geographic area based on the location of where the products were shipped or services rendered are as follows:

2019	Americas	Europe	Asia	Total	% of Total
	(Amounts in thousands)				
TS	$ 67,228	$ 3,285	$ 646	$ 71,159	90 %
HPP	5,294	771	1,837	7,902	10 %
Total	$ 72,522	$ 4,056	$ 2,483	$ 79,061	100 %
% of Total	92 %	5 %	3 %	100 %	
2018					
TS	$ 52,034	$ 9,059	$ 1,344	$ 62,437	86 %
HPP	8,424	1,266	789	10,479	14 %
Total	$ 60,458	$ 10,325	$ 2,133	$ 72,916	100 %
% of Total	83 %	14 %	3 %	100 %	

Substantially all Americas amounts are United States.

Long-lived assets by geographic location at years ended September 30, 2019 and 2018 were as follows:

	September 30, 2019	September 30, 2018
	(Amounts in thousands)	
North America	$ 1,270	$ 844
Europe	3	3
Totals	$ 1,273	$ 847

Deferred tax assets by geographic location at years ended September 30, 2019 and 2018 were as follows:

	September 30, 2019	September 30, 2018
	(Amounts in thousands)	
North America	$ 1,946	$ 1,895
Europe	—	—
Totals	$ 1,946	$ 1,895

The following table lists customers from which the Company derived revenues in excess of 10% of total revenues for the years ended years ended September 30, 2019 and 2018.

	For the years ended			
	September 30, 2019		September 30, 2018	
	Amount	% of Revenues	Amount	% of Revenues
	(Amounts in millions)			
Customer A	$ 3.8	5 %	$ 7.5	10 %
Customer B	$ 10.2	13 %	$ 1.1	3 %

In addition, accounts receivable from Customer A totaled approximately $0.3 million, or 1%, and approximately $1.1 million, or 9%, of total consolidated accounts receivable as of September 30, 2019 and September 30, 2018, respectively. Accounts receivable and long term receivable from Customer B totaled approximately $7.4 million, or 36%, and approximately $0.2 million, or 2%, of total consolidated accounts receivable as of September 30, 2019 and September 30, 2018, respectively. We believe that the Company is not exposed to any significant credit risk with respect to the accounts receivable with these customers as of September 30, 2019. No other customers accounted for 10% or more of total consolidated accounts receivable as of September 30, 2019.

Same-store sales at company-operated restaurants increased1.7% in 2019 compared with 2018 primarily due to menu price increases and favorable product mix, partially offset by a decline in transactions. The following table summarizes the increases (decreases) in company-operated same-store sales:

	2019 vs. 2018
Transactions	(1.4)%
Average check (1)	3.1 %
Change in same-store sales	1.7 %

(1) Includes price increases of approximately 2.5% in 2019.

Food and packaging costs as a percentage of company restaurant sales increased to29.0% in 2019 from 28.8% a year ago due primarily to changes in product mix and higher costs for ingredients, partially offset by menu price increases. Commodity costs increased by approximately 2% compared to a year ago, due to increases across the majority of our ingredient costs.

Payroll and employee benefit costs as a percentage of company restaurant sales increased to29.7% in 2019 compared with 28.8% a year ago due primarily to higher average wages resulting from wage inflation and a highly competitive labor market, and a change in the mix of restaurants due to refranchising.

Occupancy and other costs decreased $21.2 million in 2019 compared to the prior year primarily driven by a decrease in the average number of restaurants impacting occupancy and other costs by approximately $23.0 million. As a percentage of company restaurant sales, occupancy and other costs decreased to 15.0% from 16.0% a year ago due primarily to refranchising and a decrease in maintenance and repair expenses, partially offset by higher costs for insurance, information technology, and delivery fees at the restaurants we continue to operate.

Franchise Operations

The following table presents franchise revenues and costs in each fiscal year and other information we believe is useful in analyzing the change in franchise operating results (*dollars in thousands*):

	2019	2018
Franchise rental revenues	$ 272,815	$ 259,047
Royalties	163,047	155,939
Franchise fees and other	6,764	6,646
Franchise royalties and other	169,811	162,585
Franchise contributions for advertising and other services	170,674	—
Total franchise revenues	$ 613,300	$ 421,632
Franchise occupancy expenses (excluding depreciation and amortization)	$ 166,584	$ 158,319
Franchise support and other costs	12,110	11,593
Franchise advertising and other services expenses	178,093	—
Total franchise costs	$ 356,787	$ 169,912
Franchise costs as a % of total franchise revenues	58.2%	40.3%
Average number of franchise restaurants	2,081	2,028
% increase	2.6%	
Franchised restaurant sales	$ 3,167,920	$ 3,018,067
Franchise restaurant AUV	$ 1,523	$ 1,488
Increase in franchise-operated same-store sales	1.3%	0.1%
Royalties as a percentage of total franchise restaurant sales	5.1%	5.2%

Franchise rental revenues increased $13.8 million, or 5.3%, in 2019 versus a year ago due primarily to an increase in the number of franchised restaurants and, to a lesser extent, an increase in franchise same-store sales. The increase in the number of restaurants leased or subleased from the Company due to our refranchising strategy, contributed additional rental revenues of $12.4 million in 2019.

SIGNIFICANT ASSUMPTIONS

We used the following key assumptions to measure the post-employment benefit obligations and the net benefit plans cost for the DB pension plans and OPEB plans. These assumptions are long-term, which is consistent with the nature of post-employment benefit plans.

	DB PENSION PLANS AND OPEB PLANS	
FOR THE YEAR ENDED DECEMBER 31	2019	2018
Post-employment benefit obligations		
Discount rate	**3.1%**	3.8%
Rate of compensation increase	**2.25%**	2.25%
Cost of living indexation rate [1]	**1.6%**	1.6%
Life expectancy at age 65 (years)	**23.2**	23.1

(1) *Cost of living indexation rate is only applicable to DB pension plans.*

	DB PENSION PLANS AND OPEB PLANS	
FOR THE YEAR ENDED DECEMBER 31	2019	2018
Net post-employment benefit plans cost		
Discount rate	**4.0%**	3.7%
Rate of compensation increase	**2.25%**	2.25%
Cost of living indexation rate [1]	**1.6%**	1.6%
Life expectancy at age 65 (years)	**23.1**	23.2

(1) *Cost of living indexation rate is only applicable to DB pension plans.*

The weighted average duration of the post-employment benefit obligation is 14 years.

We assumed the following trend rates in healthcare costs:

- an annual increase in the cost of medication of 6.5% for 2019 decreasing to 4.0% over 20 years
- an annual increase in the cost of covered dental benefits of 4%
- an annual increase in the cost of covered hospital benefits of 3.7%
- an annual increase in the cost of other covered healthcare benefits of 4%

Assumed trend rates in healthcare costs have a significant effect on the amounts reported for the healthcare plans.

The following table shows the effect of a 1% change in the assumed trend rates in healthcare costs.

EFFECT ON POST-EMPLOYMENT BENEFITS – INCREASE/(DECREASE)	1% INCREASE	1% DECREASE
Total service and interest cost	4	(3)
Post-employment benefit obligations	110	(95)

SENSITIVITY ANALYSIS

The following table shows a sensitivity analysis of key assumptions used to measure the net post-employment benefit obligations and the net post-employment benefit plans cost for our DB pension plans and OPEB plans.

		IMPACT ON NET POST-EMPLOYMENT BENEFIT PLANS COST FOR 2019 – INCREASE/(DECREASE)		IMPACT ON POST-EMPLOYMENT BENEFIT OBLIGATIONS AT DECEMBER 31, 2019 – INCREASE/(DECREASE)	
	CHANGE IN ASSUMPTION	INCREASE IN ASSUMPTION	DECREASE IN ASSUMPTION	INCREASE IN ASSUMPTION	DECREASE IN ASSUMPTION
Discount rate	0.5%	(75)	69	(1,728)	1,944
Life expectancy at age 65	1 year	38	(39)	945	(972)

POST-EMPLOYMENT BENEFIT PLAN ASSETS

The investment strategy for the post-employment benefit plan assets is to maintain a diversified portfolio of assets invested in a prudent manner to maintain the security of benefits.

The following table shows the target allocations for 2019 and the allocation of our post-employment benefit plan assets at December 31, 2019 and 2018.

	WEIGHTED AVERAGE TARGET ALLOCATION	TOTAL PLAN ASSETS FAIR VALUE	
ASSET CATEGORY	2019	DECEMBER 31, 2019	DECEMBER 31, 2018
Equity securities	**0%–40%**	**22%**	20%
Debt securities	**60%–100%**	**62%**	64%
Alternative investments	**0%–50%**	**16%**	16%
Total		**100%**	100%

20. SUPPLEMENTAL CASH FLOW INFORMATION

Supplemental cash flow information and non-cash investing and financing activities are as follows for the years ended December 31,:

	2019	2018	2017
Supplemental cash flow information:			
Cash paid for interest	$ 750.2	$ 789.7	$ 712.1
Cash paid for income taxes (net of refunds of $11.2, $25.0 and $20.7, respectively)	147.5	163.9	136.5
Non-cash investing and financing activities:			
(Decrease) increase in accounts payable and accrued expenses for purchases of property and equipment and construction activities	(21.0)	8.3	34.0
Purchases of property and equipment under finance leases, perpetual easements and capital leases	81.3	57.8	54.8
Fair value of debt assumed through acquisitions	329.8	—	—
Acquisition of Commercialization Rights (1)	—	24.8	—
Conversion of third-party debt to equity	—	—	48.2
Debt financed acquisition of communication sites	—	54.2	—

(1) Related to the note extinguishment with TV Azteca, S.A. de C.V. in 2018.

21. BUSINESS SEGMENTS

The Company's primary business is leasing space on multitenant communications sites to wireless service providers, radio and television broadcast companies, wireless data providers, government agencies and municipalities and tenants in a number of other industries. This business is referred to as the Company's property operations.

During the fourth quarter of 2019, as a result of recent acquisitions, including the Eaton Towers Acquisition, and changes to its organizational structure, the Company reviewed and changed its reportable segments to divide its EMEA property segment into two segments: Africa property and Europe property. Prior to this revision, the Company operated in five business segments: (i) U.S. property, (ii) Asia property (iii) EMEA property, (iv) Latin America property and (v) services. The change is consistent with how the chief operating decision maker reviews financial performance and operating and business management strategies for each of the six segments. The change in reportable segments had no impact on the Company's consolidated financial statements for any periods. Historical financial information included in this Annual Report on Form 10-K has been adjusted to reflect the change in reportable segments.

As of December 31, 2019, the Company's property operations consisted of the following:

- U.S.: property operations in the United States;
- Asia: property operations in India;
- Africa: property operations in Burkina Faso, Ghana, Kenya, Niger, Nigeria, South Africa and Uganda;
- Europe: property operations in France and Germany; and
- Latin America: property operations in Argentina, Brazil, Chile, Colombia, Costa Rica, Mexico, Paraguay and Peru.

The Company's services segment offers tower-related services in the United States, including AZP and structural analysis, which primarily support its site leasing business, including the addition of new tenants and equipment on its sites. The services segment is a strategic business unit that offers different services from, and requires different resources, skill sets and marketing strategies than, the property operating segments.

The accounting policies applied in compiling segment information below are similar to those described in note 1. Among other factors, in evaluating financial performance in each business segment, management uses segment gross margin and segment operating profit. The Company defines segment gross margin as segment revenue less segment operating expenses excluding stock-based compensation expense recorded in costs of operations; Depreciation, amortization and accretion; Selling, general, administrative and development expense; and Other operating expenses. The Company defines segment operating profit as segment gross margin less Selling, general, administrative and development expense attributable to the segment, excluding stock-based compensation expense and corporate expenses. For reporting purposes, for periods through September 30, 2018, the Latin America property segment gross margin and segment operating profit also included Interest income (expense), TV

utilization. The Company has approximately $33.7 of tax-effected state net operating loss carryforwards (without regard to federal benefit of state). Some of these net operating loss carryforwards have an indefinite carryforward period, and those that do not will begin to expire in 2020 if not utilized. The state net operating loss carryforwards are primarily related to Florida and New Jersey, but the Company has smaller net operating losses in various other states. The Company has approximately $65.6 of tax-effected foreign net operating loss carryforwards. Some of these net operating loss carryforwards have an indefinite carryforward period, and those that do not will begin to expire in 2020 if not utilized. The foreign net operating loss carryforwards increased from 2018 to 2019 primarily due to the recognition of a discrete tax benefit of $41.0 in connection with a foreign restructuring plan allowing the future realization of net operating losses. Additionally, the Company has $5.0 of U.S. federal and state research and development tax credit carryforwards (without regard to federal benefit of state). Some of these research and development credit carryforwards have an indefinite carryforward period, and those that do not will begin to expire in 2020 if not utilized.

As of December 31, 2019, the Company determined that a total valuation allowance of $36.3 was necessary to reduce U.S. federal and state deferred tax assets by $15.4 and foreign deferred tax assets by $20.9, where it was more likely than not that all of such deferred tax assets will not be realized. As of December 31, 2019, the Company believes it is more likely than not that the remaining net deferred tax assets will be realized based on the Company's estimates of future taxable income and any applicable tax-planning strategies within various tax jurisdictions.

The Company recognizes in the Consolidated Financial Statements only those tax positions determined to be "more likely than not" of being sustained upon examination based on the technical merits of the positions. A reconciliation of the beginning and ending amount of unrecognized tax benefits is as follows:

	2019	2018	2017
Beginning balance	**$ 63.6**	$ 52.2	$ 38.7
Additions for tax positions of prior periods	**2.9**	2.4	24.8
Additions for tax positions of the current period	**4.2**	6.9	4.2
Additions due to acquisitions	**1.9**	4.4	—
Reductions for tax positions of prior periods	**(0.3)**	(0.4)	(11.2)
Reductions attributable to settlements with taxing authorities	**—**	—	(1.5)
Reductions attributable to lapses of applicable statute of limitations	**(2.5)**	(1.9)	(2.8)
Ending balance	**$ 69.8**	$ 63.6	$ 52.2

The total amount of unrecognized tax benefits that, if recognized, would impact the effective tax rate is $68.2. Interest and penalties related to unrecognized tax benefits were $1.8 in 2019 and are classified as a component of income tax expense. Accrued interest and penalties were $8.7 at December 31, 2019 and $6.9 at December 31, 2018. During the next twelve months, it is reasonably possible that the unrecognized tax benefits may decrease by a net $6.3, mainly due to anticipated statute of limitations lapses in various jurisdictions.

The Company and its subsidiaries are subject to examinations for U.S. federal income tax as well as income tax in various state, city and foreign jurisdictions. The Company's federal income tax returns for 2016 through the current period remain open to examination and the relevant state, city and foreign statutes vary. The Company does not expect the assessment of any significant additional tax in excess of amounts reserved.

The Tax Act was signed into U.S. law on December 22, 2017. The Tax Act contains provisions which impact the Company's income taxes including a reduction in the U.S. federal corporate income tax rate from 35% to 21%, a one-time deemed mandatory repatriation tax imposed on all undistributed foreign earnings, and the introduction of a modified territorial taxation system.

The SEC released Staff Accounting Bulletin No. 118 ("SAB 118") on December 22, 2017 to provide guidance where the accounting under ASC 740, *Income Taxes*, is incomplete for certain income tax effects of the Tax Act upon issuance of financial statements for the reporting period in which the Tax Act was enacted. SAB 118 provides that if a company could determine a reasonable estimate, that estimate should be reported as a provisional amount and adjusted during a measurement period. If a company is unable to determine a reasonable estimate, no related provisional amounts would be recorded until a reasonable estimate can be determined, within the measurement period. The measurement period extends until all necessary information has been obtained, prepared, and analyzed, but no longer than 12-months from the date of enactment of the Tax Act.

The Company intends to distribute all historical unremitted foreign earnings up to the amount of excess foreign cash, as well as all future foreign earnings that can be repatriated without incremental U.S. federal tax cost. Any remaining outside basis differences relating to the Company's investments in foreign subsidiaries are no longer expected to be material and will be indefinitely reinvested.

(8) LONG-TERM DEBT

On September 23, 2016, Roper entered into a five-year $2.5 billion unsecured credit facility, as amended December 2, 2016, (the "2016 Facility") with JPMorgan Chase Bank, N.A., as administrative agent, and a syndicate of lenders, which replaced its previous $1.85 billion unsecured credit facility dated as of July 27, 2012, as amended as of October 28, 2015 (the "2012 Facility"). The 2016 Facility comprises a five year $2.5 billion revolving credit facility, which includes availability of up to $150.0 for letters of credit. Roper may also, subject to compliance with specified conditions, request term loans or additional revolving credit commitments in an aggregate amount not to exceed $500.0. At December 31, 2019, there were $0.0 of outstanding borrowings under the 2016 Facility.

Note 22. Supplemental Financial Information

Cash paid for income taxes amounted to $77.6 million, $25.9 million and $48.4 million during fiscal 2019, 2018 and 2017, respectively. Cash paid for interest on borrowings amounted to $347.9 million, $85.3 million and $82.5 million during fiscal 2019, 2018 and 2017, respectively.

A summary of additions and deductions related to the valuation allowance for deferred tax asset accounts for the years ended March 31, 2019, 2018 and 2017 follows (amounts in millions):

	Balance at Beginning of Year	Additions Charged to Costs and Expenses	Additions Charged to Other Accounts	Deductions	Balance at End of Year
Valuation allowance for deferred tax assets:					
Fiscal 2019	$ 204.5	$ 16.2	$ 175.8	$ (64.4)	$ 332.1
Fiscal 2018	$ 210.1	$ 36.2	$ —	$ (41.8)	$ 204.5
Fiscal 2017	$ 161.8	$ 15.2	$ 37.6	$ (4.5)	$ 210.1

A summary of additions and deductions related to the allowance for doubtful accounts for the years ended March 31, 2019, 2018 and 2017 follows (amounts in millions):

	Balance at Beginning of Year	Additions Charged to Costs and Expenses	Deductions [1]	Balance at End of Year
Allowance for doubtful accounts:				
Fiscal 2019	$ 2.2	$ —	$ (0.2)	$ 2.0
Fiscal 2018	$ 2.1	$ 0.2	$ (0.1)	$ 2.2
Fiscal 2017	$ 2.5	$ 0.2	$ (0.6)	$ 2.1

[1] Deductions represent uncollectible accounts written off, net of recoveries.

Accumulated Other Comprehensive Income

The following tables present the changes in the components of accumulated other comprehensive income (AOCI) for the years ended March 31, 2019 and 2018:

	Unrealized Holding Gains (Losses) Available-for-sale Securities	Minimum Pension Liability	Foreign Currency	Total
Balance at March 31, 2018	$ 1.9	$ (10.1)	$ (9.4)	$ (17.6)
Impact of change in accounting principle	(1.7)	—	—	(1.7)
Opening Balance as of April 1, 2018	0.2	(10.1)	(9.4)	(19.3)
Other comprehensive loss before reclassifications	(5.6)	2.9	(5.3)	(8.0)
Amounts reclassified from accumulated other comprehensive income (loss)	5.6	1.0	—	6.6
Net other comprehensive income (loss)	—	3.9	(5.3)	(1.4)
Balance at March 31, 2019	$ 0.2	$ (6.2)	$ (14.7)	$ (20.7)

On a constant currency basis, total research and development expenses were flat in fiscal 2019, as lower employee related expenses including lower variable compensation were offset by an increase in stock-based compensation expenses .

General and Administrative Expenses: General and administrative expenses primarily consist of personnel related expenditures for IT, finance, legal and human resources support functions; and professional services fees.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
General and administrative (1)	$ 1,093	-1%	2%	$ 1,102
Stock-based compensation	172	-5%	-5%	180
Total expenses	$ 1,265	-1%	1%	$ 1,282
% of Total Revenues	3%			3%

(1) Excluding stock-based compensation

Excluding the effects of currency rate fluctuations, total general and administrative expenses increased in fiscal 2019 compared to fiscal 2018 primarily due to increased professional services fees.

Amortization of Intangible Assets: Substantially all of our intangible assets were acquired through our business combinations. We amortize our intangible assets over, and monitor the appropriateness of, the estimated useful lives of these assets. We also periodically review these intangible assets for potential impairment based upon relevant facts and circumstances. Note 6 of Notes to Consolidated Financial Statements included elsewhere in this Annual Report has additional information regarding our intangible assets and related amortization.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Developed technology	$ 857	13%	14%	$ 758
Cloud services and license support agreements and related relationships	712	-3%	-3%	731
Other	120	-9%	-9%	131
Total amortization of intangible assets	$ 1,689	4%	4%	$ 1,620

Amortization of intangible assets increased in fiscal 2019 compared to fiscal 2018 primarily due to additional amortization from intangible assets, which primarily included developed technology that we acquired in connection with our recent acquisitions, partially offset by a reduction in expenses associated with certain of our intangible assets that became fully amortized.

Acquisition Related and Other Expenses: Acquisition related and other expenses consist of personnel related costs and stock-based compensation for transitional and certain other employees, integration related professional services, and certain business combination adjustments including certain adjustments after the measurement period has ended and certain other operating items, net. Stock-based compensation expenses included in acquisition related and other expenses resulted from unvested restricted stock-based awards and stock options assumed from acquisitions whereby vesting was accelerated generally upon termination of the employees pursuant to the original terms of those restricted stock-based awards and stock options.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Transitional and other employee related costs	$ 49	3%	4%	$ 48
Stock-based compensation	—	-100%	-100%	1
Professional fees and other, net	16	373%	426%	3
Business combination adjustments, net	(21)	*	*	—
Total acquisition related and other expenses	$ 44	-15%	-13%	$ 52

* Not meaningful

17. Personnel expenses

Personnel expenses can be broken down as follows:

€ million	2017/18	**2018/19**
Wages and salaries	2,173	**2,264**
Social security expenses, expenses for post-employment benefits and related employee benefits	552	**552**
thereof for post-employment benefits	(43)	**(41)**
	2,725	**2,816**

Wages and salaries include expenses relating to restructuring measures and severance payments of €23 million (2017/18: €19 million). The variable remuneration rose from €52 million in financial year 2017/18 to €81 million in financial year 2018/19. Wages and salaries also include expenses for long-term remunerationcomponents totalling €7 million (2017/18: €16 million).

Annual average number of group employees in the continuing segment:

Number of employees by headcount	2017/18	**2018/19**
Blue collar/white collar	103,072	**99,843**
Apprentices/trainees	1,840	**1,811**
	104,912	**101,654**

This includes an absolute number of 16,902 (2017/18: 17,245) part-time employees. The number of employees working outside of Germany stood at 81,607 (2017/18: 84,425). This includes 80,660 blue and white collar employees (2017/18: 83,498). In addition, 947 trainees were employed abroad (2017/18: 927).

18. Other taxes

The other taxes (for example property tax, motor vehicle tax, excise tax and transaction tax) have the following effects on the income statement:

€ million	2017/18	**2018/19**
Other taxes	**79**	**79**
thereof from cost of sales	(1)	**(1)**
thereof from selling expenses	(65)	**(62)**
thereof from general administrative expenses	(13)	**(16)**

The decrease in internal financing revenue was due to lower average asset balances, partially offset by higher asset yields. The decrease in external financing revenue reflects the wind down of the OEM IT commercial financing operations.

Sales of used equipment represented 43.4 percent and 48.5 percent of Global Financing's revenue for the years ended December 31, 2019 and 2018, respectively. The decrease in 2019 was due to a lower volume of internal used equipment sales. The gross profit margin on used sales was 52.2 percent and 54.2 percent for the years ended December 31, 2019 and 2018, respectively. The decrease in the gross profit margin was driven by lower margins on internal used equipment sales.

Global Financing pre-tax income decreased 22.5 percent year to year primarily driven by a decrease in gross profit ($422 million), partially offset by a decrease in total expense ($115 million), which was mainly driven by a decline in IBM shared expenses in line with the segment's performance, a lower provision for credit losses and a gain from the sale of certain commercial financing capabilities in the first quarter of 2019.

The decrease in return on equity from 2018 to 2019 was primarily due to lower net income. Refer to page 45 for the details of the after-tax income and return on equity calculations.

Geographic Revenue

In addition to the revenue presentation by reportable segment, we also measure revenue performance on a geographic basis.

($ in millions)

For the year ended December 31:	2019	2018	Yr.-to-Yr. Percent Change	Yr.-to-Yr. Percent Change Adjusted for Currency	Yr.-to-Yr. Percent Change Excluding Divested Businesses And Adjusted for Currency
Total revenue	$77,147	$79,591	(3.1)%	(1.0)%	0.2%
Americas	$36,274	$36,994	(1.9)%	(1.1)%	0.8%
Europe/Middle East/Africa	24,443	25,491	(4.1)	0.4	1.3
Asia Pacific	16,430	17,106	(4.0)	(3.0)	(2.5)

Total revenue of $77,147 million in 2019 decreased 3.1 percent year to year as reported (1 percent adjusted for currency), but increased 0.2 percent excluding divested businesses and adjusted for currency.

Americas revenue decreased 1.9 percent as reported (1 percent adjusted for currency), but grew 1 percent excluding divested businesses and adjusted for currency. Within North America, the U.S. decreased 2.4 percent and Canada increased 4.0 percent as reported (6 percent adjusted for currency). Latin America declined as reported but grew adjusted for currency. Within Latin America, Brazil declined 4.8 percent as reported, but was flat adjusted for currency.

EMEA revenue decreased 4.1 percent as reported, but was essentially flat adjusted for currency and increased 1 percent excluding divested businesses and adjusted for currency. As reported, the U.K., France and Italy decreased 2.9 percent, 4.1 percent and 1.3 percent, respectively, but grew 1 percent, 1 percent and 4 percent, respectively, adjusted for currency. Germany decreased 7.9 percent as reported and 3 percent adjusted for currency. The Middle East and Africa region decreased 3.5 percent as reported and 2 percent adjusted for currency.

Asia Pacific revenue decreased 4.0 percent as reported (3 percent adjusted for currency) and 2 percent excluding divested businesses and adjusted for currency. Japan increased 2.3 percent as reported and 1 percent adjusted for currency. Australia decreased 17.3 percent as reported and 11 percent adjusted for currency. China decreased 13.4 percent as reported and 11 percent adjusted for currency and India decreased 8.1 percent as reported and 5 percent adjusted for currency.

Total Expense and Other (Income)

($ in millions)

For the year ended December 31:	2019	2018	Yr.-to-Yr. Percent/ Margin Change*
Total consolidated expense and other (income)	$26,322	$25,594	2.8%
Non-operating adjustments			
Amortization of acquired intangible assets	(764)	(437)	74.8
Acquisition-related charges	(409)	(16)	NM
Non-operating retirement-related (costs)/income	(615)	(1,572)	(60.9)
Operating (non-GAAP) expense and other (income)	$24,533	$23,569	4.1%
Total consolidated expense-to-revenue ratio	34.1%	32.2%	2.0 pts.
Operating (non-GAAP) expense-to-revenue ratio	31.8%	29.6%	2.2 pts.

* 2019 results were impacted by Red Hat purchase accounting and acquisition-related activity.

NM—Not meaningful

The following Red Hat-related expenses were included in 2019 total consolidated expense and other (income), with no corresponding expense in the prior-year: Red Hat operational spending, interest expense from debt issuances to fund the acquisition and other acquisition-related activity, including: amortization of acquired intangible assets, retention and legal and advisory fees associated with the transaction.

to our revenue contracts. We did not identify any material differences resulting from applying the new requirements to our revenue contracts. In addition, we did not identify any significant changes to our business processes, systems, and controls to support recognition and disclosure requirements under the new guidance. We adopted the provisions of Topic 606 in fiscal 2019 utilizing the modified retrospective method. We recorded a $0.5 million cumulative effect adjustment, net of tax, to the opening balance of fiscal 2019 retained earnings, a decrease to receivables of $7.6 million, an increase to inventories of $2.8 million, an increase to prepaid expenses and other current assets of $6.9 million, an increase to other accrued liabilities of $1.4 million, and an increase to other noncurrent liabilities of $0.2 million. The adjustments primarily related to the timing of recognition of certain customer charges, trade promotional expenditures, and volume discounts.

The effect of the changes made to our Consolidated Balance Sheet as of May 26, 2019 for the adoption of Topic 606 was as follows:

	As Reported	Adjustments	Balances without Adoption of Topic 606
Current assets			
Receivables, less allowance for doubtful accounts	$ 831.7	$ 8.7	$ 840.4
Inventories	1,571.7	(3.1)	1,568.6
Prepaid expenses and other current assets	93.8	(16.6)	77.2
Current liabilities			
Other accrued liabilities	691.6	(1.1)	690.5
Other noncurrent liabilities	1,951.8	(2.5)	1,949.3

The effect of the changes made to our Consolidated Statement of Earnings for the adoption of Topic 606 was as follows:

	Fiscal 2019		
	As Reported	Adjustments	Balances without Adoption of Topic 606
Net sales	$ 9,538.4	$ 15.5	$ 9,553.9
Cost of goods sold	6,885.4	24.5	6,909.9
Income from continuing operations before income taxes and equity method investment earnings	823.3	(9.0)	814.3

In January 2016, the FASB issued ASU 2016-01, *Financial Instruments-Overall: Recognition and Measurement of Financial Assets and Financial Liabilities*, which addresses certain aspects of recognition, measurement, presentation, and disclosure of financial instruments. The effective date for this standard is for fiscal years beginning after December 31, 2017. We adopted this ASU in fiscal 2019. The adoption of this guidance did not have a material impact to our consolidated financial statements.

In August 2016, the FASB issued ASU 2016-15, *Statement of Cash Flows: Classification of Certain Cash Receipts and Cash Payments*, which clarifies how companies present and classify certain cash receipts and cash payments in the statement of cash flows. We adopted this ASU retrospectively in fiscal 2019. The adoption of this guidance did not have a material impact to our consolidated financial statements.

In November 2016, the FASB issued ASU 2016-18, *Statement of Cash Flows: Restricted Cash*, which provides amendments to current guidance to address the classifications and presentation of changes in restricted cash in the statement of cash flows. We adopted this ASU retrospectively in fiscal 2019. The adoption of this guidance did not have a material impact to our consolidated financial statements.

In January 2017, the FASB issued ASU 2017-01, *Business Combinations: Clarifying the Definition of a Business*, which provides a new framework for determining whether transactions should be accounted for as acquisitions (or disposals) of assets or businesses. We adopted this ASU prospectively in fiscal 2019. The adoption of this guidance did not have a material impact to our consolidated financial statements.

In March 2017, the FASB issued ASU 2017-07, *Retirement Benefits: Improving the Presentation of Net Periodic Pension Cost and Net Periodic Postretirement Benefit Cost*, which requires companies to present the service cost component of net

We experienced a gross margin percentage increase in our Americas segment due to productivity improvements, partially offset by unfavorable impacts from pricing and product mix. The unfavorable product mix impact was driven by products within the Infrastructure Platforms product category. Our gross margin in this segment also benefited from the sale of our lower margin SPVSS business during the second quarter of fiscal 2019.

Product gross margin in our EMEA segment increased due to productivity improvements, partially offset by negative impacts from pricing and mix.

The APJC segment gross margin percentage decrease was due to negative impacts from pricing, partially offset by productivity improvements and favorable product mix. Lower service gross margin also contributed to the decrease in the gross margin in this geographic segment.

The gross margin percentage for a particular segment may fluctuate, and period-to-period changes in such percentages may or may not be indicative of a trend for that segment.

Research and Development ("R&D"), Sales and Marketing, and General and Administrative ("G&A") Expenses

R&D, sales and marketing, and G&A expenses are summarized in the following table (in millions, except percentages):

	Years Ended			2019 vs. 2018	
	July 27, 2019	July 28, 2018	July 29, 2017	Variance in Dollars	Variance in Percent
Research and development	**$ 6,577**	$ 6,332	$ 6,059	$ 245	4%
Percentage of revenue	***12.7%***	*12.8%*	*12.6%*		
Sales and marketing	**9,571**	9,242	9,184	329	4%
Percentage of revenue	***18.4%***	*18.7%*	*19.1%*		
General and administrative	**1,827**	2,144	1,993	(317)	(15)%
Percentage of revenue	***3.5%***	*4.3%*	*4.2%*		
Total	**$ 17,975**	$ 17,718	$ 17,236	$ 257	1%
Percentage of revenue	***34.6%***	*35.9%*	*35.9%*		

R&D Expenses

R&D expenses increased due to higher headcount-related expenses and, to a lesser extent, higher acquisition-related costs, higher contracted services and higher discretionary spending.

We continue to invest in R&D in order to bring a broad range of products to market in a timely fashion. If we believe that we are unable to enter a particular market in a timely manner with internally developed products, we may purchase or license technology from other businesses, or we may partner with or acquire businesses as an alternative to internal R&D.

Sales and Marketing Expenses

Sales and marketing expenses increased due to higher headcount-related expenses, higher discretionary spending and, to a lesser extent, higher contracted services and higher acquisition-related costs, partially offset by lower share-based compensation expense.

G&A Expenses

G&A expenses decreased due to a benefit from the $400 million litigation settlement with Arista Networks and lower contracted services, partially offset by higher discretionary spending and higher headcount-related expenses.

Effect of Foreign Currency

In fiscal 2019, foreign currency fluctuations, net of hedging, decreased the combined R&D, sales and marketing, and G&A expenses by approximately $233 million, or 1.3%, compared with fiscal 2018. In fiscal 2018, foreign currency fluctuations, net of hedging, increased the combined R&D, sales and marketing, and G&A expenses by approximately $93 million, or 0.5%, compared with fiscal 2017.

The primary components of the deferred tax assets and liabilities are as follows, for the periods indicated below:

	June 30, 2019	June 30, 2018
Deferred tax assets		
Non-capital loss carryforwards	$ 161,119	$ 129,436
Capital loss carryforwards	155	417
Undeducted scientific research and development expenses	137,253	123,114
Depreciation and amortization	683,777	829,369
Restructuring costs and other reserves	17,845	17,202
Deferred revenue	53,254	62,726
Other	59,584	57,461
Total deferred tax asset	$ 1,112,987	$ 1,219,725
Valuation Allowance	$ (77,328)	$ (80,924)
Deferred tax liabilities		
Scientific research and development tax credits	$ (14,482)	$ (13,342)
Other	(72,599)	(82,668)
Deferred tax liabilities	$ (87,081)	$ (96,010)
Net deferred tax asset	$ 948,578	$ 1,042,791
Comprised of:		
Long-term assets	1,004,450	1,122,729
Long-term liabilities	(55,872)	(79,938)
	$ 948,578	$ 1,042,791

We believe that sufficient uncertainty exists regarding the realization of certain deferred tax assets that a valuation allowance is required. We continue to evaluate our taxable position quarterly and consider factors by taxing jurisdiction, including but not limited to factors such as estimated taxable income, any historical experience of losses for tax purposes and the future growth of OpenText.

The aggregate changes in the balance of our gross unrecognized tax benefits (including interest and penalties) were as follows:

Unrecognized tax benefits as of July 1, 2017	$ 174,530
Increases on account of current year positions	6,483
Increases on account of prior year positions	17,794
Decreases due to settlements with tax authorities	—
Decreases due to lapses of statutes of limitations	(20,995)
Unrecognized tax benefits as of June 30, 2018	$ 177,812
Increases on account of current year positions	25,642
Increases on account of prior year positions	15,024
Decreases due to settlements with tax authorities	—
Decreases due to lapses of statutes of limitations	(9,236)
Unrecognized tax benefits as of June 30, 2019	$ 209,242

Included in the above tabular reconciliation are unrecognized tax benefits of $11.2 million relating to deferred tax assets in jurisdictions in which these deferred tax assets are offset with valuation allowances. The net unrecognized tax benefit excluding these deferred tax assets is approximately $198.1 million as of June 30, 2019 (June 30, 2018—$167.2 million).

We recognize interest expense and penalties related to income tax matters in income tax expense. For the year ended June 30, 2019, 2018 and 2017, we recognized the following amounts as income tax-related interest expense and penalties:

CAPITAL EXPENDITURES

Capital expenditures include costs associated with acquiring property, plant and equipment and placing it into service. The telecommunications business requires extensive and continual investments, including investment in new technologies and the expansion of capacity and geographical reach. Expenditures related to the acquisition of spectrum licences and additions to right-of-use assets are not included in capital expenditures and do not factor into the calculation of free cash flow or capital intensity. See "Managing Our Liquidity and Financial Resources", "Key Performance Indicators", and "Non-GAAP Measures and Related Performance Measures" for more information.

Capital expenditures are significant and have a material impact on our cash flows; therefore, our management teams focus on planning, funding, and managing them. We believe this measure best reflects our cost of property, plant and equipment in a given period and is a simpler measure for comparing between periods.

(In millions of dollars, except capital intensity)	Years ended December 31		
	2019	2018	% Chg
Capital expenditures [1]			
Wireless	**1,320**	1,086	22
Cable	**1,153**	1,429	(19)
Media	**102**	90	13
Corporate	**232**	185	25
Capital expenditures [1]	**2,807**	2,790	1
Capital intensity [2]	**18.6%**	18.5%	0.1 pts

[1] Includes additions to property, plant and equipment net of proceeds on disposition, but does not include expenditures for spectrum licences or additions to right-of-use assets.

[2] As defined. See "Key Performance Indicators".

WIRELESS

The increase in capital expenditures in Wireless this year was a result of investments made to upgrade our wireless network to continue delivering reliable performance for our customers. We continued augmenting our existing LTE network with 4.5G technology investments that are also 5G-ready to prepare for the commercial launch of 5G in select markets in early 2020.

In 2019, we acquired spectrum licences for $1,731 million, which is not included in the table above. See "Managing Our Liquidity and Financial Resources".

CABLE

The decrease in capital expenditures in Cable this year was a result of lower purchases of customer premise equipment and lower investments related to the initial launch of Ignite TV. We have continued upgrading our network infrastructure with additional fibre deployments, including increasing our fibre-to-the-home and fibre-to-the-curb distribution. These upgrades will lower the number of homes passed per node and incorporate the latest technologies to help deliver more bandwidth and an even more reliable customer experience as we progress in our Connected Home roadmap.

MEDIA

The increase in capital expenditures this year was a result of higher investments in renovations at various Toronto Blue Jays facilities, partially offset by lower investment in our broadcast and IT infrastructure and the sale of our publishing business.

CORPORATE

The increase in Corporate capital expenditures this year was a result of higher investments in IT and our various real estate facilities this year, including the impact of $25 million of proceeds from the sale of certain assets last year.

CAPITAL INTENSITY

Capital intensity this year was in line with 2018.